Getting Started Linux

Getting Started Tutorial

This page contains the most common steps for setting up and getting started with your Engaging account. We provide this page as a convenient reference to get started. Each system has its own in-depth documentation which can be found on the Engaging System page.

Getting an Account

If you don't already have an account, login into the ORCD OnDemand Portal https://orcd-ood.mit.edu using your MIT kerberos credentials. The system will then be prompted to create your account automatically. Wait a few minutes for the system to create all the pieces for your account before submitting your first job. On rare occasions this could take up to an hour.

Logging In

The first thing you should do when you get a new account is verify that you can log in. Engaging provides multiple ways to log in, including both ssh and OnDemand.

Terminal with SSH

Log into Engaging with the following command in a terminal window. Replace `USERNAME` below with your MIT Kerberos username:

```
ssh USERNAME@orcd-login.mit.edu
```

You will be prompted for your Kerberos password and then for Duo two-factor authentication.

If you are using older Centos 7 nodes you can use one of the Centos 7 login nodes instead:

- `orcd-vlogin001`
- `orcd-vlogin002`
- `orcd-vlogin003`
- `orcd-vlogin004`

See Logging in with SSH for more information.

OnDemand

You can log into OnDemand Web Portal with the link: https://orcd-ood.mit.edu. For full detailed instructions please see the OnDemand Documentation.

Shared HPC Clusters

Engaging is a shared HPC cluster. You are sharing this resources with a number of other researchers, staff, and students, so it is important that you read this page and use the system as intended.

Being a cluster, there are several machines connected together with a network. We refer to these as **nodes**. Most nodes in the cluster are referred to as **compute nodes**, this is where the computation is done on the system (where you will run your code). When you ssh into the system you are on a special purpose node called the **login node**. The login node, as its name suggests, is where you log in and is for editing code and files, installing packages and software, downloading data, and starting jobs to run your code on one of the compute nodes.

Each job is started using a piece of software called the **scheduler**, which you can think of as a resource manager. You let it know what resources you need and what you want to run, and the scheduler will find those resources and start your job on them. When your job completes those resources are relinquished. The scheduler is what ensures that no two jobs are using the same resources, so it is very important not to run anything unless it is submitted properly through the scheduler.

Software and Packages

The first thing you may want to do is make sure the system has the software and packages you need. We have installed a lot of software and packages on the system already, even though it may not be immediately obvious that it is there. There are several pages in the "Software" section of this site. We recommend reading through both the Overview and Modules pages, and then select the additional pages most relevant to you.

If you are ever unsure if we have a particular software, and you cannot find it, please send us an email and ask before you spend a lot of time trying to install it. If we have it, we can point you to it, provide advice on how to use it, and if we don't have it we can often give pointers on how to install

it. Further, if a lot of people request the same software, we may consider adding it to the system image.

Linux Command Line

Engaging runs Linux, so much of what you do on the cluster involves the Linux command line. That doesn't mean you have to be a Linux expert to use the system! However the more you can get comfortable with the Linux command line and a handful of basic commands, the easier using the system will be. If you are already familiar with Linux, feel free to skip this section, or skim as a refresher.

Most Linux commands deal with **directories** and **files**. A **directory**, synonymous to a folder, contains files and other directories. The list of directories that lead to a particular directory or file is called its **path**. In Linux, directories on a path are separated by forward slashes `/`. It is also important to note that everything in Linux is case sensitive, so a file `myScript.sh` is not the same as the file `myscript.sh`. When you first log in, you are in your **home directory**. Your home directory is where you can put all the code and data you need to run your job. Your home directory is not accessible to other users, so if you need a space to share files with other users, let us know and we can make a shared **group directory** for you.

The path to your home directory on Engaging is `/home/<USERNAME>`, where `<USERNAME>` is your username. The character `~` is also shorthand for your home directory in any Linux commands.

Anytime after you start typing a Linux command you can press the "Tab" button your your keyboard. This called tab-complete, and will try to autocomplete what you are typing. This is particularly helpful when typing out long directory paths and file names. Pressing "Tab" once will complete if there is a single completion, pressing it twice will list all potential completions. It is a bit difficult to explain in text, but you can try it out yourself and watch the short demonstration here.

Finally, click on the box below for a list of Linux Commands. If you are new to Linux try them out for yourself at the command line.

Common Linux Commands

- Creating, navigating and viewing directories:
  - `pwd` : tells you the full path of the directory you are currently in
  - `mkdir dirname` : creates a directory with the name "dirname"
  - `cd dirname` : change directory to directory "dirname"
    - `cd ../` : takes you up one level
  - `ls` : lists the files in the directory
    - `ls -a` : lists all files including hidden files
    - `ls -l` : lists files in "long format" including ownership and date of last update
    - `ls -t` : lists files by date stamp, most recently updated file first
    - `ls -tr` : lists files by dates stamp in reverse order, most recently updated file is listed last (this is useful if you have a lot of files, you want to know which file you changed last and the list of files results in a scrolling window)
    - `ls dirname` : lists the files in the directory "dirname"
- Viewing files
  - `more filename` : shows the first part of a file, hitting the space bar allows you to scroll through the rest of the file, q will cause you to exit out of the file.
  - `less filename` : allows you to scroll through the file, forward and backward, using the arrow keys.
  - `tail filename` : shows the last 10 lines of a file (useful when you are monitoring a log file or output file to see that the values are correct)
    - `tail <number> filename` : show you the last <number> lines of a file.
    - `tail -f filename` : shows you new lines as they are written to the end of the file. Press CMD+C or Control+C to exit. This is helpful to monitor the log file of a batch job.
- Copying, moving, renaming, and deleting files
  - `mv filename dirname` : moves filename to directory dirname.
    - `mv filename1 filename2` : moves filename1 to filename2, in essence renames the file. The date and time are not changed by the mv command.
  - `cp filename dirname` : copies to directory dirname.
    - `cp filename1 filename2` : copies filename1 to filename2. The date stamp on filename2 will be the date/time that the file was moved
    - `cp -r dirname1 dirname2` : copies directory dirname1 and its contents to dirname2.

- `rm filename` : removes (deletes) the file

Filesystems

Everyone on Engaging gets three spaces to store files: home, pool, and scratch. Each of these have a different purpose, size, and characteristics. You can read more about the standard user filesystems on the General Use Filesystems page.

Additional shared storage can be rented through ORCD. See our Storage Services page for more information.

Transferring Files

One of the first tasks is to get your code, data, and any other files you need into your home directory on the system. If your code is in GitHub you can use git commands on the system to clone your repository to your home directory. You can also transfer your files to one of your Engaging directories from your computer by using: - OnDemand File browser - Globus - The `scp` or `rsync` commands in your terminal

Read the page on Transferring Files to learn more about how to transfer what you need to Engaging.

Running your First Job

At this point you may want to do a test-run of your code. You always want to start small in your test runs, so you should choose a small example that tests the functionality of what you would ultimately like to run on the system. If your test code is serial and runs okay on a moderate personal laptop or desktop you can request an interactive session to run your code in by executing the command:

```
# Requesting a single core for an interactive job for 1 hour
salloc -t 01:00:00 -p mit_normal
```

After you run this command you will be on a compute node and you can do a test-run of your code. This command will allocate one core to your job. If your test code is multithreaded or parallel, uses a lot of memory, or requires a GPU you should request additional resources as needed. Not requesting the resources you will be using can negatively impact others on the system.

Review the "Running Jobs" section of this site. We recommend reading through both the Overview and Requesting Resources pages, and then select any additional pages most relevant to you.

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